

4.1.2 Database Management System (DBMS)

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 02 of 15

4.1.2 Database Management System (DBMS)

LEARN IT

A **DBMS** is software that lets you create, manage and interact with databases. It sits between the user and the stored files. You never edit raw data files by hand — you talk to the DBMS (Access, MySQL, Oracle, SQLite) and it handles storage, security and consistency.

Advantages you must be able to list:

- **Less redundancy.** The department name is stored once, not in every student row.
- **Consistency.** Change a phone number once; every screen that uses it shows the new number.
- **Security.** Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- **Integrity.** Rules stop duplicate roll numbers and empty primary keys.

WORKED EXAMPLE

In a paper register, Ali's address is written on the admission page, the fee page and the sports page. If he moves house, two of the three pages stay wrong. In a DBMS, Address lives in one **Student** table. Every form and report reads that single field.

اردو میں سمجھیں

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔ اس سے تکرار کم ہوتی ہے، ڈیٹا ایک جیسا رہتا ہے، صرف مجاز افراد دیکھ سکتے ہیں، اور غلط اندراج رک جاتی ہے۔

CHECK YOURSELF

1. Name two DBMS programs.
2. Which advantage stops two students sharing the same roll number?

Answers: 1. MS Access, MySQL (Oracle, SQLite also). 2. Integrity (unique / primary-key constraint).

Next lecture: 4.2 Database Components (Tables, Records, Fields)